Decision Science - Solved Question Papers

MBA Semester I (2024 Pattern) | Paper 1: [6430]-506 | Paper 2: [6380]-5006

PAPER 1 : [6430]-506 | GC-06 Decision Science (DS-506 MJ)

Q1) Solve any five (2 marks each) - all eight solved

a) Explain PERT.

PERT (Programme Evaluation and Review Technique) is a network technique used to plan and control projects where activity times are uncertain. Three time estimates are used: optimistic (a), most likely (m) and pessimistic (b). Expected time te = (a + 4m + b)/6 and variance = $((b - a)/6)^2$. It gives the critical path and the probability of finishing the project on time.

b) What is Saddle Point?

In a two-person zero-sum game, a saddle point is the cell of the pay-off matrix where the **maximin value (largest of row minima) equals the minimax value (smallest of column maxima)**. That cell is the value of the game, and it is the smallest in its row and largest in its column. Game with a saddle point is solved by pure strategies.

c) What is Hungarian Method?

It is a method to solve the assignment problem (assigning n jobs to n workers at minimum cost). Steps: subtract row minimum from each row, subtract column minimum from each column, cover all zeros with minimum number of lines, and if lines = n, make assignment on zeros; otherwise subtract the smallest uncovered element from uncovered elements and add it to elements at intersection of lines, and repeat.

d) What is Pure strategy game?

A pure strategy game is a game in which each player selects one particular course of action every time the game is played, with certainty. It exists when the game has a saddle point, so the optimal strategy of each player is a single fixed strategy.

e) Explain method to obtain feasible solution in transportation problem.

An initial basic feasible solution (satisfying all supply and demand conditions) can be obtained by: **North-West Corner Method (NWCM)** - start allocation from top-left cell; **Least Cost Method (LCM)** - allocate first to the cell with lowest cost; and **Vogel's Approximation Method (VAM)** - allocate using row and column penalties (difference between two lowest costs). VAM generally gives the solution closest to the optimal. The number of allocated cells should be m + n - 1.

f) What is unbalanced transportation problem?

A transportation problem in which **total supply is not equal to total demand**. To solve it, it is converted into a balanced problem: if supply is more, a dummy destination is added; if demand is more, a dummy source is added. The cost of the dummy row/column is taken as zero.

g) Explain CPM.

CPM (Critical Path Method) is a network technique for planning and scheduling projects in which the activity durations are known with certainty. It identifies the **critical path**, which is the longest path in the network and decides the minimum project duration. Activities on this path have zero float, so any delay in them delays the project.

h) What is EMV criterion in decision making under risk?

Expected Monetary Value (EMV) criterion is used when probabilities of states of nature are known. EMV of an act = sum of (pay-off x probability of the state). The act with the **highest EMV** (for profit) or the lowest EMV (for cost) is selected.

Q2) Solve any two out of three (5 marks each) - all three solved

a) Explain role of quantitative techniques in management decision making process.

Quantitative techniques (linear programming, transportation, assignment, game theory, network analysis, decision theory, queuing, simulation) apply mathematical models to managerial problems. Their role:

- **Scientific decision making:** replace guesswork by data and models.
- **Optimum use of resources:** LP gives the best product mix to maximise profit or minimise cost.
- **Problem structuring:** a complex problem is converted into a model so alternatives can be compared.
- **Planning and control:** PERT/CPM for project scheduling; transportation for distribution cost reduction.
- **Decision under risk and uncertainty:** decision trees, EMV and game theory help in selecting the best strategy.
- **Time and cost saving:** quick evaluation of many alternatives, especially with computer software.

b) With suitable example elaborate difference between CPM and PERT.

Differentiate between PERT and CPM.

Basis	PERT	CPM
Full form	Programme Evaluation and Review Technique	Critical Path Method
Origin	US Navy, 1958 (Polaris missile project)	DuPont and Remington Rand, 1957
Time estimate	Three estimates: optimistic (a), most likely (m), pessimistic (b); te = (a + 4m + b)/6	Single deterministic time estimate
Nature	Probabilistic, event oriented	Deterministic, activity oriented
Used for	Research and development, new projects with uncertain times	Repetitive projects like construction, maintenance with known times
Focus	Controlling time	Time-cost trade-off (crashing)
Probability	Probability of completing project by a date can be found	Not possible

Example: Constructing a building where durations are known (excavation 10 days, foundation 20 days) is handled by CPM. Developing a new software product, where the time of coding may be 5, 8 or 14 weeks, is handled by PERT using te = (5 + 4x8 + 14)/6 = 8.5 weeks.

c) How would you deal with assignment problems where: i) some assignments are prohibited, ii) objective is to be maximised, iii) not balanced, iv) multiple solutions

- **i) Prohibited assignments:** put a very large cost (M) in that cell. As M is very large, the cell will never be selected in the optimal solution.
- **ii) Maximisation:** convert to minimisation by subtracting every element from the largest element of the matrix (regret / opportunity matrix) or by multiplying all elements by -1, then apply the Hungarian method. The final value is calculated from the original matrix.
- **iii) Unbalanced:** if number of jobs is not equal to number of persons, add dummy rows or columns with zero cost to make the matrix square. Assignment to a dummy means that the job/person remains unassigned.
- **iv) Multiple (alternate) solutions:** if, in the final matrix, more than one set of independent zeros is possible (zeros are available in alternative positions), the problem has alternate optimal solutions, all with the same total cost. Management can choose any one based on other considerations.

Q3) Solve any one (10 marks) - both solved

a) Assignment of four jobs to four contractors (minimise cost)

Contractor	a	b	c	d
1	10	24	30	15
2	16	22	28	12
3	12	20	32	10
4	9	26	34	16

Step 1: Row reduction (subtract the smallest of each row: 10, 12, 10, 9)

	a	b	c	d
1	0	14	20	5
2	4	10	16	0
3	2	10	22	0
4	0	17	25	7

Step 2: Column reduction (column minima: 0, 10, 16, 0)

	a	b	c	d
1	0	4	4	5
2	4	0	0	0
3	2	0	6	0
4	0	7	9	7

Step 3: Cover all zeros by minimum lines

Lines: column a, row 2, row 3 = 3 lines, which is less than 4, so the solution is not optimal. Smallest uncovered element = 4. Subtract 4 from all uncovered elements and add 4 to the elements at intersection of lines (2a and 3a).

	a	b	c	d
1	0	0	0	1
2	8	0	0	0
3	6	0	6	0
4	0	3	5	3

Now minimum lines needed = 4, so the optimal assignment is possible. Start with a row/column having a single zero: Row 4 has only one zero (4-a). Then row 1 has zeros at b, c (a is taken); row 3 has zeros at b, d; row 2 has zeros at b, c, d. Choose 3-b, then 2-d, then 1-c.

Contractor	Job	Cost (lakhs)
1	c	30
2	d	12
3	b	20
4	a	9
Total		**71**

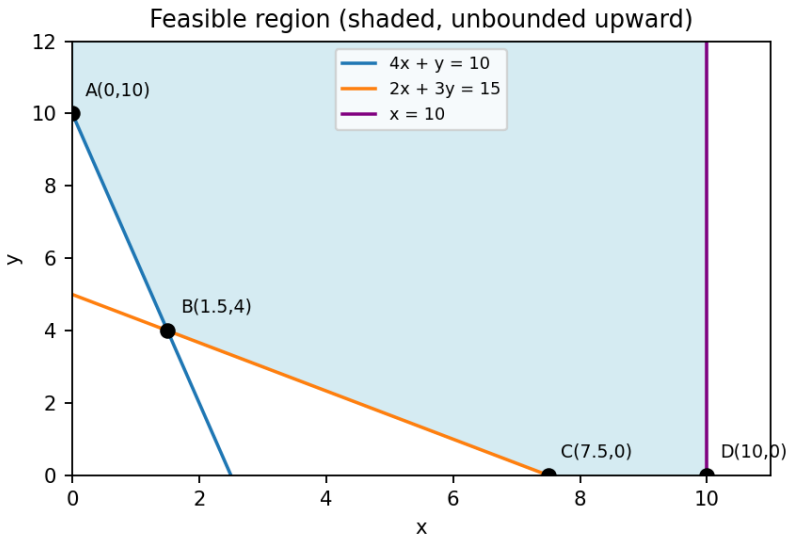
Minimum total cost = Rs. 71 lakhs. Alternative optimal assignments (e.g. 1-b, 2-c, 3-d, 4-a and 1-c, 2-b, 3-d, 4-a) also give Rs. 71 lakhs.

b) Solve LPP graphically: Minimise Z = 6x + 5y subject to 4x + y >= 10, 2x + 3y >= 15, x <= 10, x, y >= 0

Step 1 - Convert constraints into equations and find points:

- 4x + y = 10: points (0, 10) and (2.5, 0).
- 2x + 3y = 15: points (0, 5) and (7.5, 0).
- x = 10: vertical line.

Step 2 - Feasible region: Both constraints are of >= type, so the feasible region lies above both lines and to the left of x = 10. It is unbounded upward but, since Z is minimised, the minimum lies at a corner point.



Step 3 - Corner points. Intersection of 4x + y = 10 and 2x + 3y = 15: from first, y = 10 - 4x; put in second: 2x + 30 - 12x = 15, so x = 1.5 and y = 4.

Corner point	Z = 6x + 5y
A (0, 10)	0 + 50 = 50
B (1.5, 4)	9 + 20 = 29 (minimum)
C (7.5, 0)	45 + 0 = 45
D (10, 0)	60 + 0 = 60

Result: Minimum Z = 29 at x = 1.5, y = 4.

Q4) Solve any one (10 marks) - both solved

a) Game of firms A and B (find optimal strategies and value of game)

Firm A \ Firm B	B1	B2	B3	B4	Row min
A1	35	65	25	5	5
A2	30	20	15	0	0
A3	40	50	0	10	0
A4	55	60	10	15	10
Column max	55	65	25	15	

Step 1 - Saddle point test: Maximin = 10, Minimax = 15. They are not equal, so there is no saddle point and a mixed strategy is needed.

Step 2 - Dominance: Firm B wants smaller values. Column B4 (5, 0, 10, 15) is smaller than B1 (35, 30, 40, 55) and B2 (65, 20, 50, 60) in every row, so B1 and B2 are deleted. For firm A, row A1 (25, 5) dominates A2 (15, 0), and A4 (10, 15) dominates A3 (0, 10), so A2 and A3 are deleted. The reduced 2 x 2 game is:

	B3	B4
A1	25	5
A4	10	15

No saddle point (maximin 10, minimax 15). Use formulas with a = 25, b = 5, c = 10, d = 15; denominator = (a + d) - (b + c) = 40 - 15 = 25.

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p(A1) = (d - c)/25 = 5/25 = 1/5 ; p(A4) = 4/5
q(B3) = (d - b)/25 = 10/25 = 2/5 ; q(B4) = 3/5
Value V = (ad - bc)/25 = (375 - 50)/25 = 13
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Check: A's expected gain against B3 = 0.2x25 + 0.8x10 = 13; against B4 = 0.2x5 + 0.8x15 = 13. Both equal, so correct.

Answer: Firm A should use A1 (give coupons) 20% of the time and A4 (increase advertising) 80%, never A2, A3. Firm B should use B3 (maintain present strategy) 40% and B4 (increase advertising) 60%, never B1, B2. **Value of the game = 13** (net increase of 13 units in A's market share).

b) Transportation problem: initial solution by NWCM, LCM and VAM

	D1	D2	D3	D4	Supply
O1	10	20	5	7	10
O2	13	9	12	8	20
O3	4	15	7	9	30
O4	14	7	1	0	40
O5	3	12	5	19	50
Demand	60	60	20	10	150

Total supply = 150 = total demand, so the problem is balanced.

i) North-West Corner Method

Allocate from top-left: O1-D1 = 10 (O1 finished); O2-D1 = 20 (O2 finished); O3-D1 = 30 (D1 demand 60 is met and O3 is finished together); O4-D2 = 40; O5-D2 = 20 (D2 finished); O5-D3 = 20; O5-D4 = 10.

Cell	Allocation	Unit cost	Cost
O1-D1	10	10	100
O2-D1	20	13	260
O3-D1	30	4	120
O4-D2	40	7	280
O5-D2	20	12	240
O5-D3	20	5	100
O5-D4	10	19	190
Total			**1,290**

Only 7 cells are occupied against m + n - 1 = 8, so the solution is degenerate (one zero allocation is added when testing optimality). **Initial cost = Rs. 1,290.**

ii) Least Cost Method

Allocate in order of lowest cost: O4-D4 (cost 0) = 10; O4-D3 (1) = 20; O5-D1 (3) = 50; O3-D1 (4) = 10; O4-D2 (7) = 10; O2-D2 (9) = 20; O3-D2 (15) = 20; O1-D2 (20) = 10.

Cell	Allocation	Unit cost	Cost
O4-D4	10	0	0
O4-D3	20	1	20
O5-D1	50	3	150
O3-D1	10	4	40
O4-D2	10	7	70
O2-D2	20	9	180
O3-D2	20	15	300
O1-D2	10	20	200
Total			**960**

Initial cost = Rs. 960.

iii) Vogel's Approximation Method (VAM)

Step	Highest penalty	Allocation
1	Column D4 = 7 (penalties: rows 2,1,3,1,2; columns 1,2,4,7)	O4-D4 = 10 (cost 0)
2	Row O4 = 6 (rows 5,3,3,6,2; columns 1,2,4)	O4-D3 = 20 (cost 1)
3	Row O3 = 11	O3-D1 = 30 (cost 4)

Step	Highest penalty	Allocation
4	Row O1 = 10	O1-D1 = 10 (cost 10)
5	Column D1 = 10	O5-D1 = 20 (cost 3)
6	Only D2 is left (60)	O5-D2 = 30, O2-D2 = 20, O4-D2 = 10

Cell	Allocation	Unit cost	Cost
O4-D4	10	0	0
O4-D3	20	1	20
O3-D1	30	4	120
O1-D1	10	10	100
O5-D1	20	3	60
O5-D2	30	12	360
O2-D2	20	9	180
O4-D2	10	7	70
Total			**910**

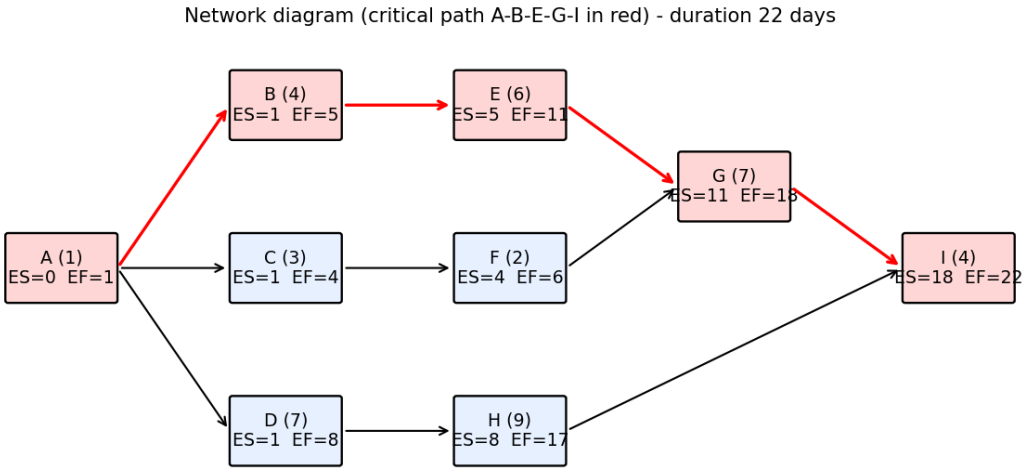
Initial cost by VAM = Rs. 910 (lowest). Comparison: NWCM 1,290; LCM 960; VAM 910. So VAM gives the best starting solution.

Q5) Solve any one (10 marks) - both solved

a) CPM project network (activities A to I)

Activity	A	B	C	D	E	F	G	H	I
Time (days)	1	4	3	7	6	2	7	9	4
Predecessor	-	A	A	A	B	C	E, F	D	G, H

i) Network diagram (with early start / early finish)



ii) Early start and early finish times; late times and float

Activity	Duration	ES	EF	LS	LF	Float
A	1	0	1	0	1	0
B	4	1	5	1	5	0
C	3	1	4	6	9	5
D	7	1	8	2	9	1
E	6	5	11	5	11	0
F	2	4	6	9	11	5

Activity	Duration	ES	EF	LS	LF	Float
G	7	11	18	11	18	0
H	9	8	17	9	18	1
I	4	18	22	18	22	0

ES = largest EF of predecessors, EF = ES + duration. For G: ES = max(EF of E = 11, EF of F = 6) = 11. For I: ES = max(EF of G = 18, EF of H = 17) = 18. Backward pass: LF of I = 22; LS = LF - duration; LF of a predecessor = smallest LS of successors.

iii) Critical path

Activities with zero float: A, B, E, G, I. **Critical path: A - B - E - G - I** and **project duration = 1 + 4 + 6 + 7 + 4 = 22 days.** (Other paths: A-D-H-I = 21 days, A-C-F-G-I = 17 days.)

iv) If duration of F is 4 days instead of 2

F then has ES = 4 and EF = 8. G still starts at 11 (because E finishes at 11, which is later than 8). Float of F reduces from 5 days to 3 days, but F is still non-critical. **The critical path (A-B-E-G-I) and project duration (22 days) do not change.** The project will be delayed only if F is delayed by more than 5 days.

b) Farmer's crop decision under uncertainty (Hurwicz alpha = 0.6, Laplace, Minimax Regret)

Rainfall	Crop A	Crop B	Crop C
High	8,000	3,500	5,000
Medium	4,500	4,500	4,900
Low	2,000	5,000	4,000

i) Hurwicz criterion (alpha = 0.6): H = 0.6 x Max + 0.4 x Min

Crop	Max	Min	Hurwicz value
A	8,000	2,000	0.6x8,000 + 0.4x2,000 = 5,600
B	5,000	3,500	0.6x5,000 + 0.4x3,500 = 4,400
C	5,000	4,000	0.6x5,000 + 0.4x4,000 = 4,600

Highest = 5,600, so **Crop A** is selected.

ii) Laplace criterion (equal probability 1/3 each)

Crop	Average pay-off
A	(8,000 + 4,500 + 2,000)/3 = 4,833.33
B	(3,500 + 4,500 + 5,000)/3 = 4,333.33
C	(5,000 + 4,900 + 4,000)/3 = 4,633.33

Highest = 4,833.33, so **Crop A** is selected.

iii) Minimax regret criterion

Regret = best pay-off in that row - given pay-off. Best pay-offs: High 8,000, Medium 4,900, Low 5,000.

Rainfall	Regret A	Regret B	Regret C
High	0	4,500	3,000
Medium	400	400	0
Low	3,000	0	1,000
Maximum regret	**3,000**	**4,500**	**3,000**

The minimum of the maximum regrets is 3,000 for both Crop A and Crop C (tie); Crop B is rejected (4,500).

Conclusion: Hurwicz and Laplace both give Crop A, and minimax regret gives A or C (tie). So the farmer should plant **Crop A**.

PAPER 2 : [6380]-5006 | 106-GC-06 Decision Science

Q1) Solve any five (2 marks each) - all eight solved

a) What is CPM?

CPM (Critical Path Method) is a project management technique with deterministic activity times. It finds the longest path (critical path) in the network, which gives the minimum project completion time, and shows the float of non-critical activities.

b) Enlist various criteria of decision making under risk.

- Expected Monetary Value (EMV) criterion
- Expected Opportunity Loss (EOL) criterion
- Expected Value of Perfect Information (EVPI)
- Decision tree analysis (also: most probable state criterion)

c) What is Mixed strategy game?

A game in which there is **no saddle point**, so each player plays his courses of action with certain probabilities (e.g. 40% strategy 1 and 60% strategy 2) instead of one fixed strategy. The probabilities are found by algebraic, arithmetic or graphical methods.

d) Explain principle of dominance.

If every pay-off of one row is greater than or equal to the corresponding pay-off of another row, then the other row is dominated and can be deleted (A, the maximiser, will never choose it). Similarly, if every element of one column is less than or equal to the corresponding element of another column, the latter column is dominated and deleted (B, the minimiser, will not choose it). It reduces the size of the game.

e) Enlist various criteria of decision making under uncertainty.

- Maximax (optimism) criterion
- Maximin (Wald / pessimism) criterion
- Minimax regret (Savage) criterion
- Hurwicz (criterion of realism)
- Laplace (equal probability) criterion

f) Explain the concept of PERT.

PERT is a network technique for projects with uncertain activity times. Three estimates (optimistic a, most likely m, pessimistic b) are used: te = (a + 4m + b)/6 and variance = $((b - a)/6)^2$. It finds the critical path, expected project duration and probability of completion on a given date.

g) What is Hungarian method?

It is an algorithm for solving the assignment problem. We reduce rows and columns by subtracting the minimum element, cover zeros by minimum number of lines, and revise the matrix until n independent zeros are obtained, which give the optimal assignment with minimum total cost.

h) Explain methods to obtain feasible solution in transportation problem.

The three methods are: **North-West Corner Method** (start allocating from top-left cell, ignores cost), **Least Cost Method** (allocate maximum to the lowest cost cell first) and **Vogel's Approximation Method** (use penalty = difference of two lowest costs in each row/column and allocate in the row/column with highest penalty). VAM gives the best initial solution.

Q2) Solve any two out of three (5 marks each) - all three solved

a) Explain importance of decision science in organizational decision making process.

- **Scientific and objective decisions:** use data and models instead of intuition.
- **Optimum allocation of scarce resources** (men, machines, money, material) using LP, transportation and assignment.
- **Handles complexity:** models simplify a complex business situation.

- **Risk and uncertainty:** decision trees, EMV, game theory support strategic decisions.
- **Project management:** PERT/CPM for time and cost control.
- **Competitive advantage:** faster, accurate and consistent decisions; helps in performance measurement and cost reduction.

b) Explain applications of Linear Programming in functional areas of management.

Area	Application
Production	Product mix to maximise profit, production scheduling, blending (feed, fuel), minimising waste (trim loss)
Marketing	Media selection (advertising budget allocation for maximum exposure), sales territory and sales force allocation
Finance	Portfolio selection, capital budgeting, working capital management, minimising cost of funds
Human resources	Manpower planning, shift scheduling, assigning employees to jobs
Logistics / Operations	Transportation and distribution (shortest route, minimum shipping cost), warehouse location, inventory

c) Differentiate between PERT and CPM.

Differentiate between PERT and CPM.

Basis	PERT	CPM
Full form	Programme Evaluation and Review Technique	Critical Path Method
Origin	US Navy, 1958 (Polaris missile project)	DuPont and Remington Rand, 1957
Time estimate	Three estimates: optimistic (a), most likely (m), pessimistic (b); te = (a + 4m + b)/6	Single deterministic time estimate
Nature	Probabilistic, event oriented	Deterministic, activity oriented
Used for	Research and development, new projects with uncertain times	Repetitive projects like construction, maintenance with known times
Focus	Controlling time	Time-cost trade-off (crashing)
Probability	Probability of completing project by a date can be found	Not possible

Q3) Solve any one (10 marks) - both solved

a) Formulate LPP model for Winner Sports advertisement problem

Step 1 - Decision variables: Let x1, x2, x3 = number of advertisements placed in magazines A, B and C respectively.

Step 2 - Exposure per advertisement = number of readers x percentage of principal buyers:

Magazine	Readers	Principal buyers %	Principal buyers (exposure per ad)	Cost per ad
A	1,00,000	15%	15,000	Rs. 5,000
B	60,000	15%	9,000	Rs. 4,500
C	40,000	7%	2,800	Rs. 4,250

Step 3 - Objective function (maximise total exposure to principal buyers):

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Maximise Z = 15,000 x1 + 9,000 x2 + 2,800 x3
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Step 4 - Constraints:

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Budget: 5,000 x1 + 4,500 x2 + 4,250 x3 <= 1,00,000
Magazine A: x1 <= 6
Magazine B: x2 >= 2
Magazine C: x3 >= 2
Non-negativity: x1, x2, x3 >= 0 (and integers, since ads are whole numbers)
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b) Assignment of machines to places (costs in Rs. thousand; '-' means not possible)

There are 4 machines and 5 places, so the problem is unbalanced. Add a dummy machine M5 with zero cost. The prohibited assignments (M2-C and M3-A) are given a very large cost M.

	A	B	C	D	E
M1	19	21	25	20	21
M2	27	24	M	25	24
M3	M	24	27	24	20
M4	22	16	20	15	16
M5 (dummy)	0	0	0	0	0

Row reduction (subtract row minima 19, 24, 20, 15, 0):

	A	B	C	D	E
M1	0	2	6	1	2
M2	3	0	M	1	0
M3	M	4	7	4	0
M4	7	1	5	0	1
M5	0	0	0	0	0

Column minima are all 0, so column reduction does not change the matrix. Zeros can be assigned in 5 independent positions (5 lines needed), so this is optimal. Row M1 has a single zero in A: M1-A. M3 has a single zero in E: M3-E. Then M2 has zero at B: M2-B. M4 has a zero at D: M4-D. Remaining dummy M5 gets place C.

Machine	Place	Cost (Rs. thousand)
M1	A	19
M2	B	24
M3	E	20
M4	D	15
M5 (dummy)	C	0 (place C is left unused)
Total		**78**

Optimal assignment cost = Rs. 78,000.

Q4) Solve any one (10 marks) - both solved

a) Find optimal strategies for A and B and the value of the game

A \ B	B1	B2	B3	Row min
A1	9	8	-7	-7
A2	3	-6	4	-6
A3	6	7	-7	-7
Column max	9	8	4	

Saddle point test: Maximin = -6, Minimax = 4. They are not equal, so there is no saddle point.

Dominance: Row A1 (9, 8, -7) is better than or equal to row A3 (6, 7, -7) in every column, so A3 is deleted. Column B2 (8, -6) is less than column B1 (9, 3), so B1 is deleted (B prefers smaller). Reduced game:

	B2	B3
A1	8	-7
A2	-6	4

Still no saddle point (maximin -6, minimax 4). With a = 8, b = -7, c = -6, d = 4: denominator = (a + d) - (b + c) = 12 + 13 = 25.

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p(A1) = (d - c)/25 = (4 + 6)/25 = 10/25 = 2/5 ; p(A2) = 3/5
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q(B2) = (d - b)/25 = (4 + 7)/25 = 11/25 ; q(B3) = 14/25
V = (ad - bc)/25 = (32 - 42)/25 = -10/25 = -2/5 = -0.4
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Check: against B2: (2/5)(8) + (3/5)(-6) = -0.4; against B3: (2/5)(-7) + (3/5)(4) = -0.4. Both equal.

Answer: Player A: (A1, A2, A3) = (2/5, 3/5, 0). Player B: (B1, B2, B3) = (0, 11/25, 14/25). **Value of game = -0.4**, so the game is slightly favourable to B.

b) Pay-offs of acts X, Y, Z - calculate VPI

State (probability)	X	Y	Z
L (0.3)	-20	-50	200
M (0.4)	200	-100	50
N (0.3)	400	600	300

Step 1: EMV of each act

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EMV(X) = 0.3(-20) + 0.4(200) + 0.3(400) = -6 + 80 + 120 = 194
EMV(Y) = 0.3(-50) + 0.4(-100) + 0.3(600) = -15 - 40 + 180 = 125
EMV(Z) = 0.3(200) + 0.4(50) + 0.3(300) = 60 + 20 + 90 = 170
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Maximum EMV = 194 (act X).

Step 2: Expected profit with perfect information (EPPI)

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Best act in L = Z (200); in M = X (200); in N = Y (600)
EPPI = 0.3(200) + 0.4(200) + 0.3(600) = 60 + 80 + 180 = 320
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Step 3: VPI (EVPI) = EPPI - Maximum EMV

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VPI = 320 - 194 = 126
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Value of perfect information = 126. The decision maker should not pay more than 126 to get perfect information about the state of nature.

Q5) Solve any one (10 marks) - both solved

a) Initial solution by NWCM, LCM and VAM

	W1	W2	W3	W4	Supply
P1	190	300	500	100	70
P2	700	300	400	600	90
P3	400	100	400	200	180
Demand	50	80	70	140	340

Supply = 70 + 90 + 180 = 340 = demand (50 + 80 + 70 + 140). Balanced.

i) North-West Corner Method

Cell	Allocation	Unit cost	Cost
P1-W1	50	190	9,500
P1-W2	20	300	6,000
P2-W2	60	300	18,000
P2-W3	30	400	12,000
P3-W3	40	400	16,000
P3-W4	140	200	28,000
Total			**89,500**

Steps: P1 has 70, W1 needs 50: allocate 50. P1 left 20 goes to W2. W2 needs 60 more from P2. P2 has 30 left for W3. W3 needs 40 more from P3, and P3 has 140 left for W4. **Cost = Rs. 89,500.**

ii) Least Cost Method

Cell	Allocation	Unit cost	Cost
P3-W2	80	100	8,000
P1-W4	70	100	7,000

Cell	Allocation	Unit cost	Cost
P3-W4	70	200	14,000
P2-W3	70	400	28,000
P3-W1	30	400	12,000
P2-W1	20	700	14,000
Total			**83,000**

Lowest cost is 100 (P1-W4 and P3-W2); allocate P3-W2 = 80 first (larger allocation) and P1-W4 = 70. Next lowest is 200 (P3-W4) = 70. Then 400 (P2-W3 = 70, P3-W1 = 30), and finally P2-W1 = 20. **Cost = Rs. 83,000.**

iii) Vogel's Approximation Method

Step	Highest penalty	Allocation
1	Column W1 = 210 (rows 90, 100, 100; columns 210, 200, 0, 100)	P1-W1 = 50
2	Row P1 = 200 and column W2 = 200 (tie; choose P1)	P1-W4 = 20
3	Column W4 = 400	P3-W4 = 120
4	Row P3 = 300	P3-W2 = 60
5	Only P2 is left (90)	P2-W2 = 20, P2-W3 = 70

Cell	Allocation	Unit cost	Cost
P1-W1	50	190	9,500
P1-W4	20	100	2,000
P3-W4	120	200	24,000
P3-W2	60	100	6,000
P2-W2	20	300	6,000
P2-W3	70	400	28,000
Total			**75,500**

Cost by VAM = Rs. 75,500. Comparison: NWCM 89,500; LCM 83,000; VAM 75,500, so VAM is the best.

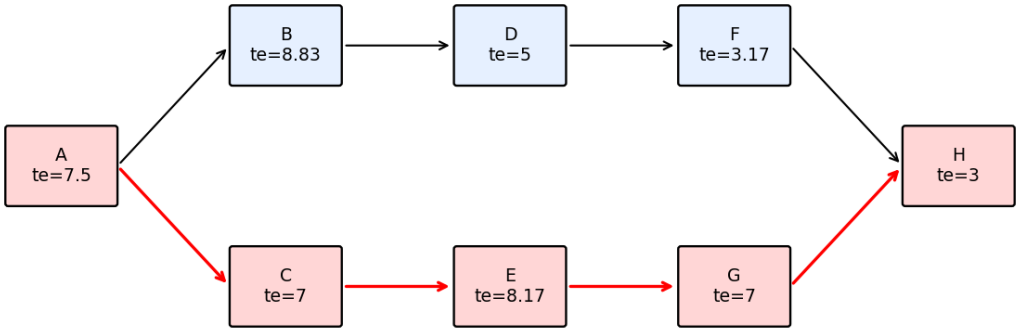
b) PERT project: expected time, network diagram and critical path

i) Expected time te = (a + 4m + b)/6 and variance ((b - a)/6)^2

Activity	Predecessor	a	m	b	te	Variance
A	-	4	7	13	(4+28+13)/6 = 7.50	2.25
B	A	6	9	11	(6+36+11)/6 = 8.83	0.69
C	A	5	7	9	(5+28+9)/6 = 7.00	0.44
D	B	3	5	7	(3+20+7)/6 = 5.00	0.44
E	C	7	8	10	(7+32+10)/6 = 8.17	0.25
F	D	2	3	5	(2+12+5)/6 = 3.17	0.25
G	E	6	7	8	(6+28+8)/6 = 7.00	0.11
H	F & G	2	3	4	(2+12+4)/6 = 3.00	0.11

ii) Network diagram with expected times

PERT network (critical path A-C-E-G-H in red)



iii) Critical path

Path	Total expected time (weeks)
A - B - D - F - H	7.5 + 8.83 + 5 + 3.17 + 3 = 27.50
A - C - E - G - H	7.5 + 7 + 8.17 + 7 + 3 = 32.67

Critical path = A - C - E - G - H (longest path). **Expected project duration = 32.67 weeks.** Variance of the critical path = 2.25 + 0.44 + 0.25 + 0.11 + 0.11 = 3.16, so standard deviation = 1.78 weeks.